

1. Problem Statement

Counting items manually in production lines or laboratory setups can be time-consuming and error-prone. This project aims to design a fully automatic item counter that displays real-time counts on a two-digit 7-segment display. Instead of using a push button for manual incrementing, an Infrared (IR) sensor module detects the passing of an object and generates a pulse. The system uses IC 555 as a clock pulse generator (triggered by the IR sensor), IC 4026 for BCD counting and 7-segment decoding, and two cascaded 4026 ICs to count from 00 to 99. A push-button reset switch resets the count to zero.

2. Literature Review

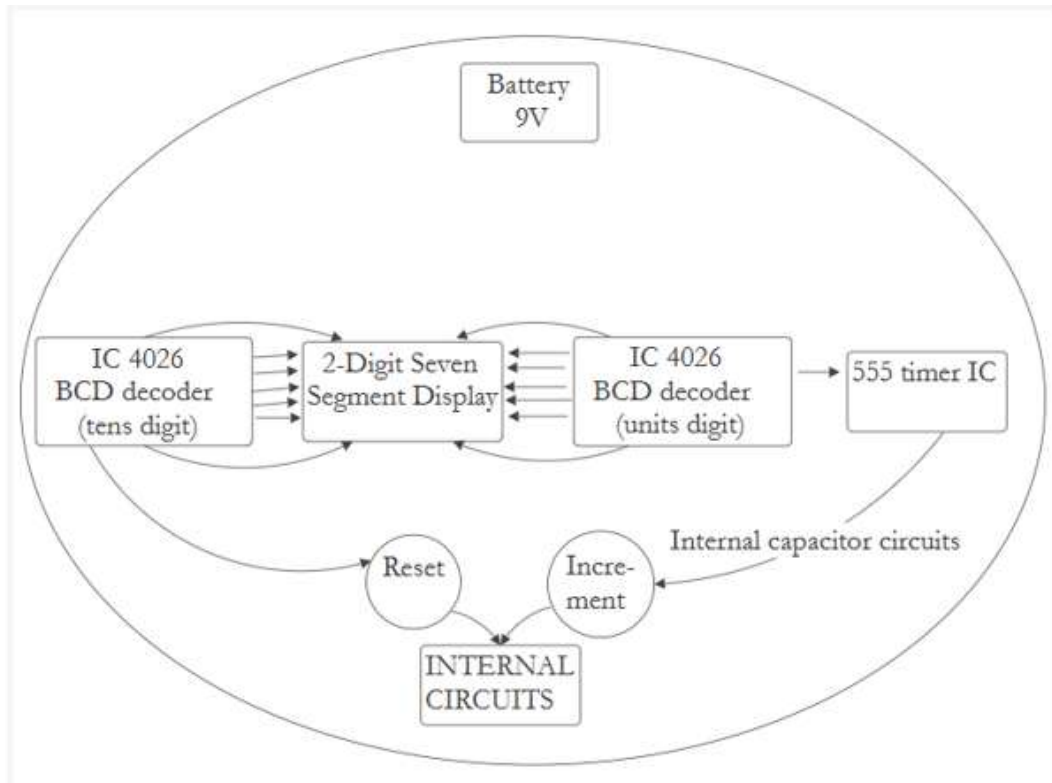
Seven-segment displays are commonly used for numerical representation in digital systems such as clocks, counters, and production line indicators. The 4026 IC is an efficient decade counter with an inbuilt 7-segment decoder, reducing external circuitry. The 555 timer IC can generate clock pulses or be used for signal conditioning. IR sensors provide reliable non-contact object detection, making them ideal for automation projects. This project combines these components to design an efficient automatic object counter.

3. Proposed Work

The proposed project is to design and implement an automatic object counter capable of displaying counts from 00 to 99 using two IC 4026 decade counters, an IR sensor module as a pulse generator, and a push-button reset circuit. Each time an object passes the IR sensor, a pulse is generated that increments the displayed count. The carry output of the first 4026 drives the clock of the second 4026 for tens counting.

4. Block Diagram

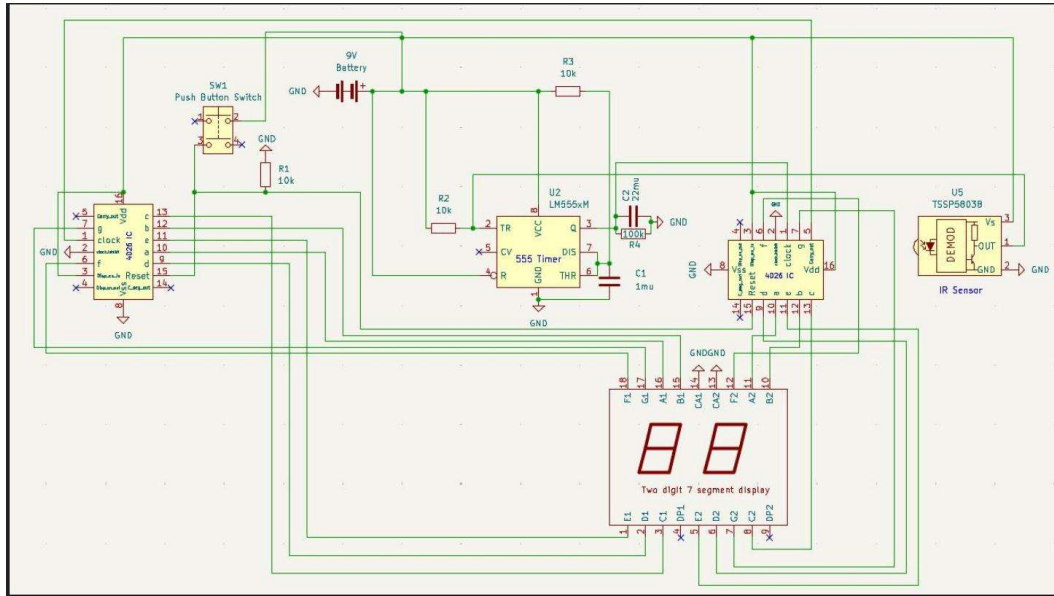
The block diagram of the automatic object counter is shown below:



The IR sensor detects the passing object and provides the input pulse to the counting section. The 4026 ICs decode the pulse sequence into readable numbers displayed on the dual 7-segment display. The reset button resets both ICs to zero.

5. Circuit Diagram

The circuit diagram for the Automatic Object Counter is shown below. The IR sensor detects object interruption and generates a pulse that triggers the 555 timer in monostable mode. The clean output pulse is used as the clock for IC 4026.



Circuit Description:

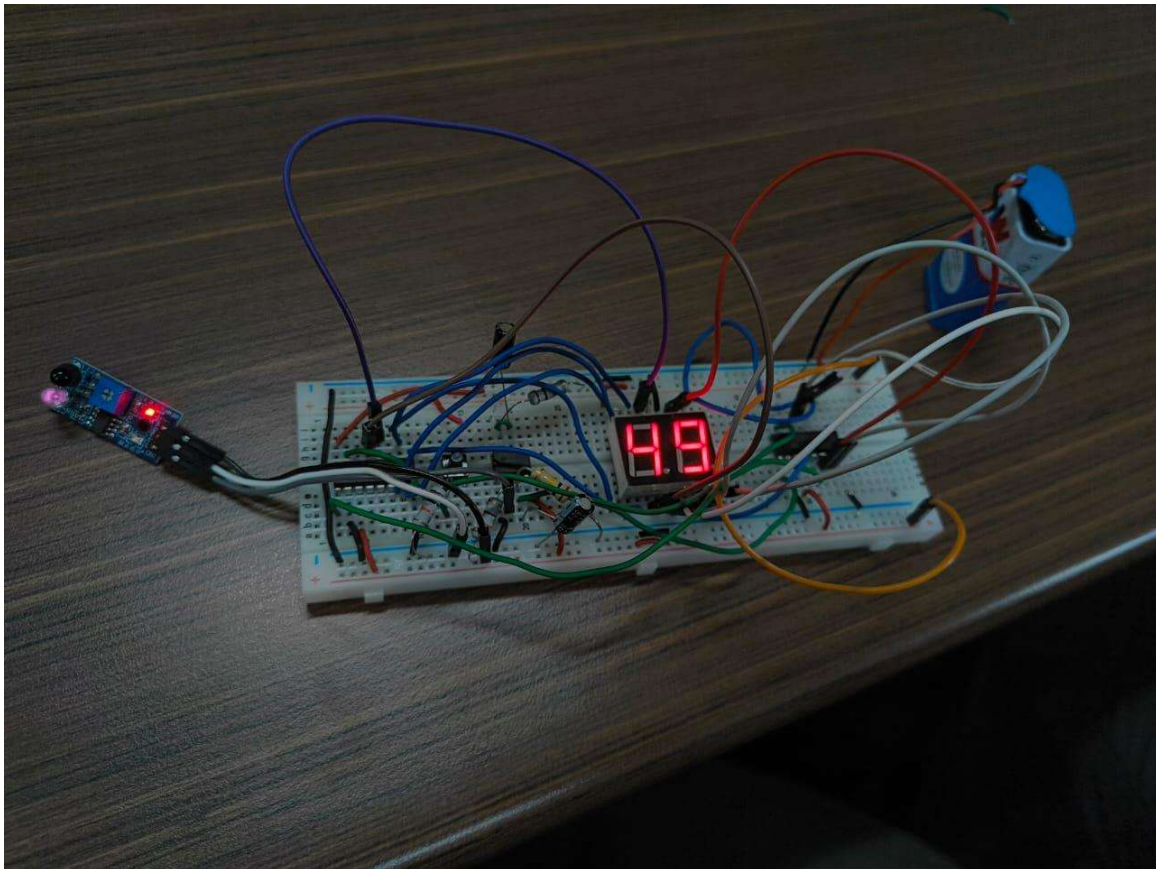
- The IR sensor provides output when an object is detected.
- 555 Timer shapes this signal to provide a clean clock pulse.
- First 4026 IC controls unit digit; carry-out triggers second 4026.
- Reset button clears both counters.
- Powered by 9V battery through suitable resistors and capacitors.

6. Working of Each Component

Component	Function
IR Sensor	Detects object presence and outputs pulse.
IC 555	Shapes IR pulse or debounces signal.
IC 4026	Counts BCD input and drives 7-segment display.
7-Segment Display	Displays numeric count (00–99).
Reset Button	Resets counter to 00.
Power Supply	Provides 9V regulated DC source.

7. Output Snapshot

Photograph of the working prototype:



8. Results and Discussion

The counter successfully increments as objects pass the IR sensor and resets upon button press. The system maintains stable performance up to approximately 2 detections per second.

9. Applications

- Production line product counters
- People counting at doorways
- Entry/exit monitoring systems
- Educational demonstration kit

10. Conclusion

The project successfully demonstrates an automatic object counter using IR sensor input and dual 4026 ICs. It eliminates the need for manual counting, improving reliability and efficiency. Future enhancements could include interfacing with microcontrollers for data logging or wireless display systems.